Proposal: Translating Handwritten Mathematical Equations into LaTeX

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1 Overview

Our goal is to build a model which performs Handwritten Mathematical Expression Recognition (HMER), specifically the generation of LaTeXwhose rendered form matches a given input image. The standard benchmark for this task, CROHME [1], is uses digital stroke paths rendered to clean images. Much less work has looked at camera photos of writing (e.g. pen, pencil, whiteboard), which come with tilted angles, shadows, variable stroke width, and background noise like notebook lines. We'll begin with an encoder-decoder architecture trained on the CROHME dataset for validation on the benchmark, then explore additional training and processing techniques to extend its performance to photographed expressions.

Encoder-Decoder Architecture Based on the literature, we intend to use a DenseNet CNN for encoding image features. The features, along with positional encodings, are fed into a transformer to generate a series of LaTeXtokens. The transformer will be bidirectionally trained using the technique described by Zhao et al [2].

Training on Photographic and Augmented Data To extend the model's effectiveness to photograph inputs, we need to train and evaluate it on more challenging data. We will use the HME100K dataset for this. However, we propose also to use CROHME to generate synthetic data with various types of adjustments such as perspective shift, distortion, noise, and stroke width variation. These will not be as realistic as the HME100K dataset, but they will allow for much easier evaluation/comparison of features the model struggles with, especially using attention visualization and other inspection techniques. This will inform further attempts to improve the model.

Preprocessing Photographs. In addition to training the model on more data, we can improve performance on photographic data by adding a preprocessing step to the inference pipeline. This could include perspective and scale correction, noise reduction, and improved color-to-grayscale conversion. Many document "scanning" or preprocessing tools exist, so we could likely achieve this by tuning an existing tool rather than building one from scratch.

2 Related Work

Deng et al. [3] introduced image-to-LaTeX with a CNN encoder and an attention decoder, and showed that attention-based models handle this task where CTC-based OCR does not. They used computer-generated images, and encountered issues with lack of attention coverage. Zhang et al. (WAP) [4] extended the technique to handwritten math and added coverage attention, which feeds the sum of past attention back into the decoder to avoid repeated or skipped symbols; we will evaluate whether the transformer effectively solves the coverage problem. Zhao et al. (BTTR) [2] replaced the RNN decoder with a transformer and trained it in both reading directions, a novel idea which notably improved ExpRate on CROHME. Our baseline follows their design. Yuan et al.(SAN) [5] decode along a syntax tree with an attention variant that sums past attention only along the tree path; for the time being, we do not intend to explore methods driven by handwritten syntax. They also released HME100K, a large, mostly camera-captured handwriting dataset, which we will use as our main data for training and testing final model performance.

3 Datasets

CROHME (2014, 2016, 2019, 2023 competition sets) is our main training and validation source: 8,836 training expressions with 101 symbol classes, and at least 3,000 test expressions. Expressions are stored online as pen trajectories (InkML), however our focus on photographed expressions means we choose to render the trajectories to offline images. The rendering step provides flexibility in scale and stroke width which we may exploit to train a more robust model.

HME100K [5] has 74,502 training and 24,607 test images with 247 symbol classes from about ten thousand writers, mostly captured by cameras. We plan to use its test images (or a subset) to evaluate on camera input. We'll use the training set to train our final model once we've validated the architecture on CROHME.

4 Plan of Activities

In general, Lon will be responsible for "back-half" functionality (image loading, DenseNet/CNN implementation) and Chris will be responsible for "front-half" (transformer implementation, inference stage, collection/normalization of results for comparison). Analysis and discussion will be shared.

We'll begin by training a working model on the CROHME dataset and comparing to existing results. Then we'll extend training with the HME100K dataset, doing evaluation against CROHME with injected irregularities so we can determine which features the model most struggles with. We'll then attempt various photograph preprocessing techniques to mitigate these issues.

By the final report we expect a working recognizer, a comparison of preprocessing and training augmentation on camera-captured input, and a clear understanding of which features the model most struggles with. If training is slower than expected on our local GPU, we will use the cluster for the longer runs and reduce the number of configurations in the camera-input study.

References

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